iEEG Session Quality Control Report

Subject: `M1012`

Session: `0`

Experiment: `MemMapOpenLoop`
Generated: 2026-08-04

Module 1: Data Completeness and Protocol Adherence

1.1 Event Count Verification

Event Type	Actual	Expected	Status
session_start	1	1	PASS ✓
trial_start	13	12, 13	PASS ✓
encoding_word_start	39	36, 39	PASS ✓
stimming	48	48	PASS ✓
distractor_phase_start	13	12, 13	PASS ✓
recognition_phase_start	13	12, 13	PASS ✓
question_1	1	1	PASS ✓

7/7 checks passed

We expect two valid session formats: with or without a practice trial.

1.2 Stimulation Delivery Validation and Trial Phasing

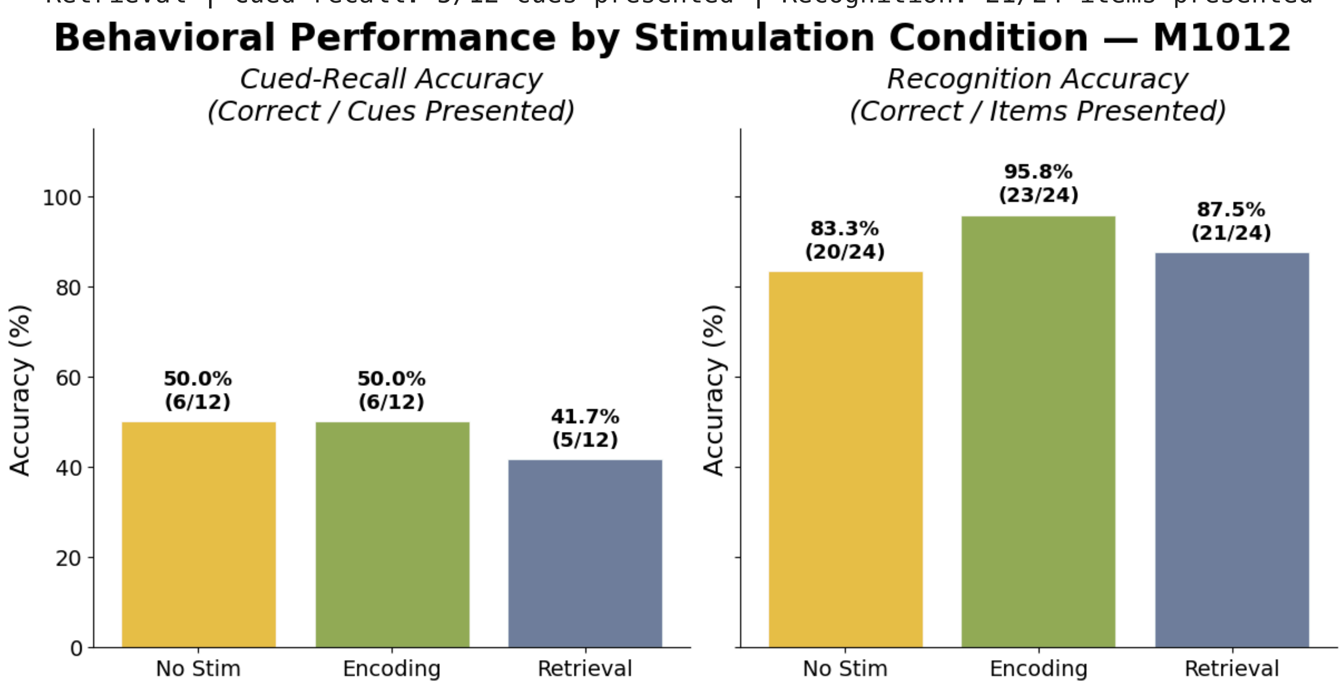
Trial	Stimulation Frequency (Hz)	Number of Encoding Stim	Number of Cue Recall Stim	Number of Recognition Stim	Stim Channel
-999	—	0	0	0	—
1	—	0	0	0	—
2	50	0	3	6	LB1-LB2
3	50	3	0	0	LB1-LB2
4	—	0	0	0	—
5	50	3	0	0	LB1-LB2
6	50	0	3	6	LB1-LB2
7	50	0	3	6	LB1-LB2
8	50	3	0	0	LB1-LB2
9	—	0	0	0	—
10	50	0	3	6	LB1-LB2
11	50	3	0	0	LB1-LB2
12	—	0	0	0	—

Module 2: Behavioral Performance Summary

```
Recognition trial types found:
  fixation_stop
  input_event
  recognition_key_pressed
  recognition_phase_start
  recognition_word_start
  recognition_word_stop
  stimming

Recognition item-start event used: recognition_word_start

======================================================================
BEHAVIORAL SUMMARY QC
======================================================================
Total non-practice trials: 12
Total cues presented: 36
Total cues correct: 17
Total recognition items presented: 72
Total recognition items correct: 64
```

Trial	Stim Condition	Cued Recall Correct	Total Cued Recall Items	Cued Recall Accuracy	Recognition Correct	Total Recognition Items	Recognition Missing	Recognition Accuracy
1	None	1	3	33.3%	6	6	0	100.0%
2	Retrieval	1	3	33.3%	6	6	0	100.0%
3	Encoding	3	3	100.0%	5	6	0	83.3%
4	None	2	3	66.7%	3	6	0	50.0%
5	Encoding	1	3	33.3%	6	6	0	100.0%
6	Retrieval	3	3	100.0%	6	6	0	100.0%
7	Retrieval	0	3	0.0%	4	6	0	66.7%
8	Encoding	1	3	33.3%	6	6	0	100.0%
9	None	3	3	100.0%	6	6	0	100.0%
10	Retrieval	1	3	33.3%	5	6	0	83.3%
11	Encoding	1	3	33.3%	6	6	0	100.0%
12	None	0	3	0.0%	5	6	0	83.3%

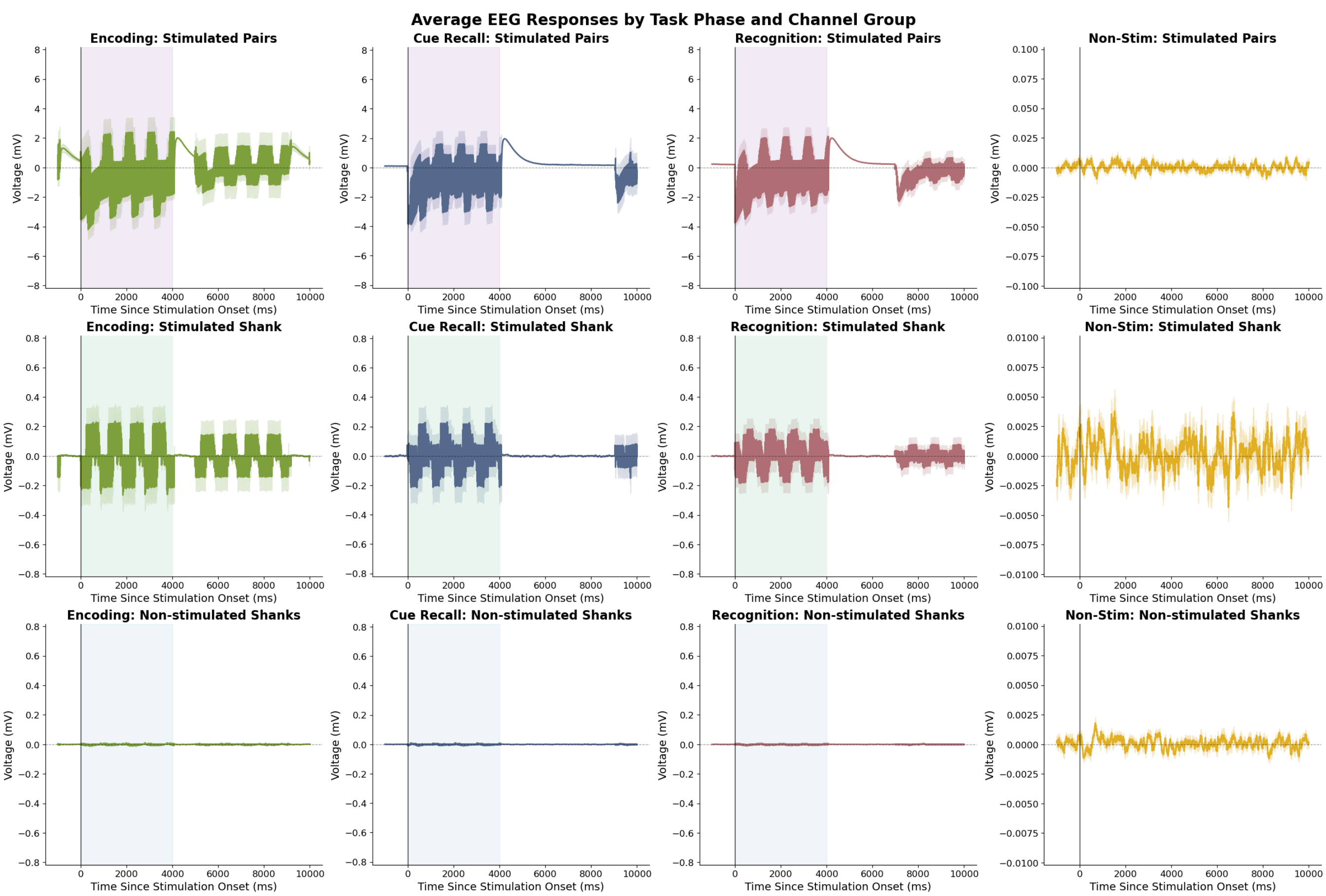
```
Pooled behavioral counts by stimulation condition:
  None      | Cued recall: 6/12 cues presented | Recognition: 20/24 items presented
  Encoding  | Cued recall: 6/12 cues presented | Recognition: 23/24 items presented
  Retrieval | Cued recall: 5/12 cues presented | Recognition: 21/24 items presented
```

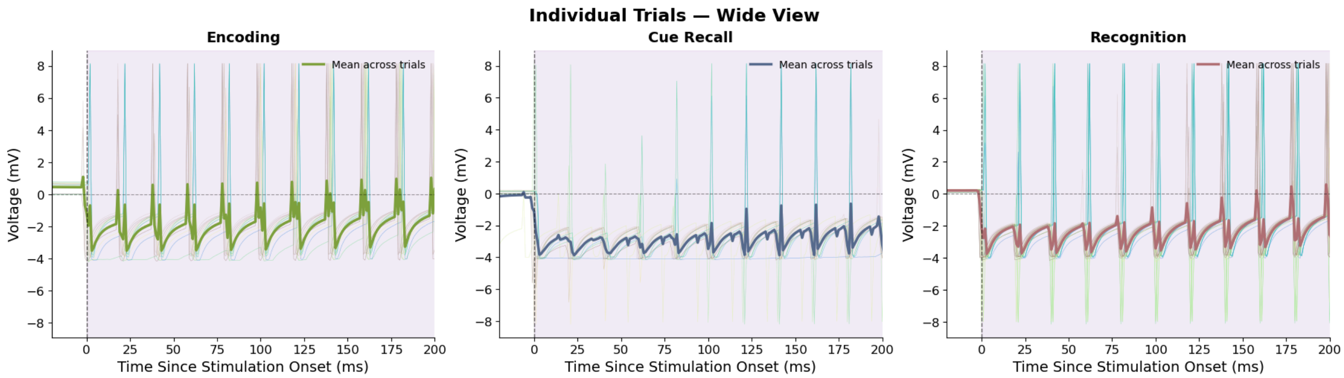


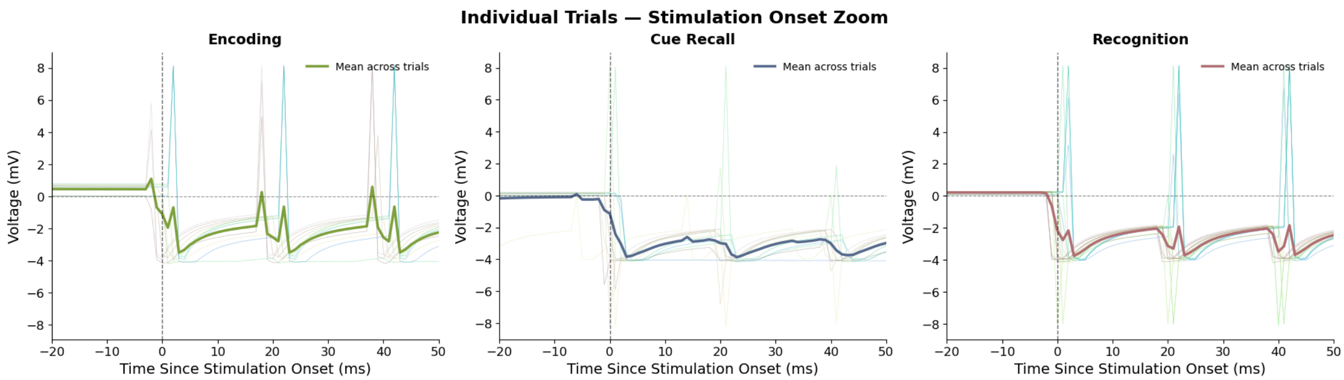
Stim Condition	Cued Recall Correct	Total Cued Recall Items	Cued Recall Accuracy	Recognition Correct	Total Recognition Items	Recognition Accuracy
None	6	12	50.0%	20	24	83.3%
Encoding	6	12	50.0%	23	24	95.8%
Retrieval	5	12	41.7%	21	24	87.5%

Module 3: Verification of Stimulation Timing, Spatial Specificity, and Frequency

3.1 EEG Stimulation Response



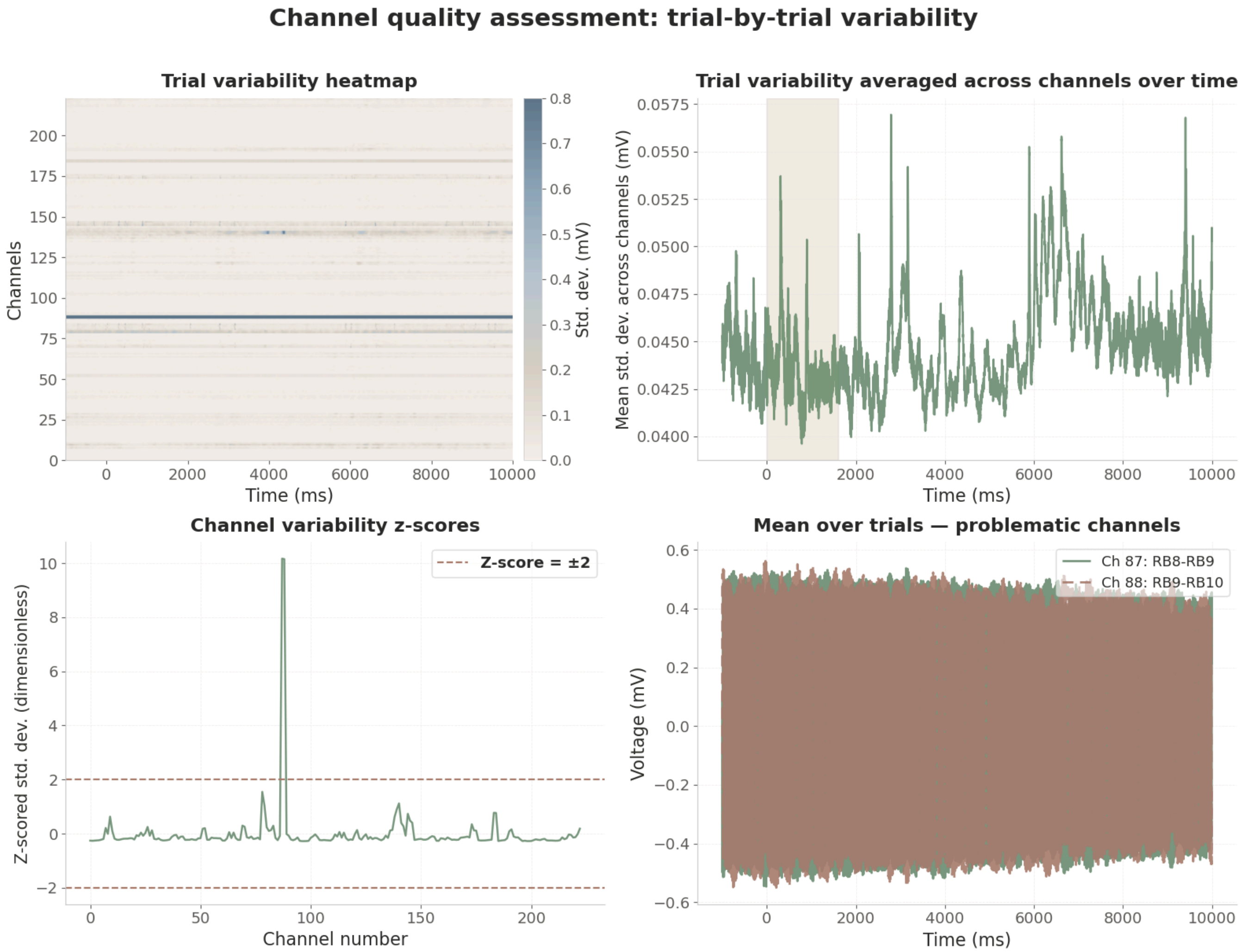




Module 4: Channel Integrity and Artifact Assessment

4.1 Trial-to-Trial Variability

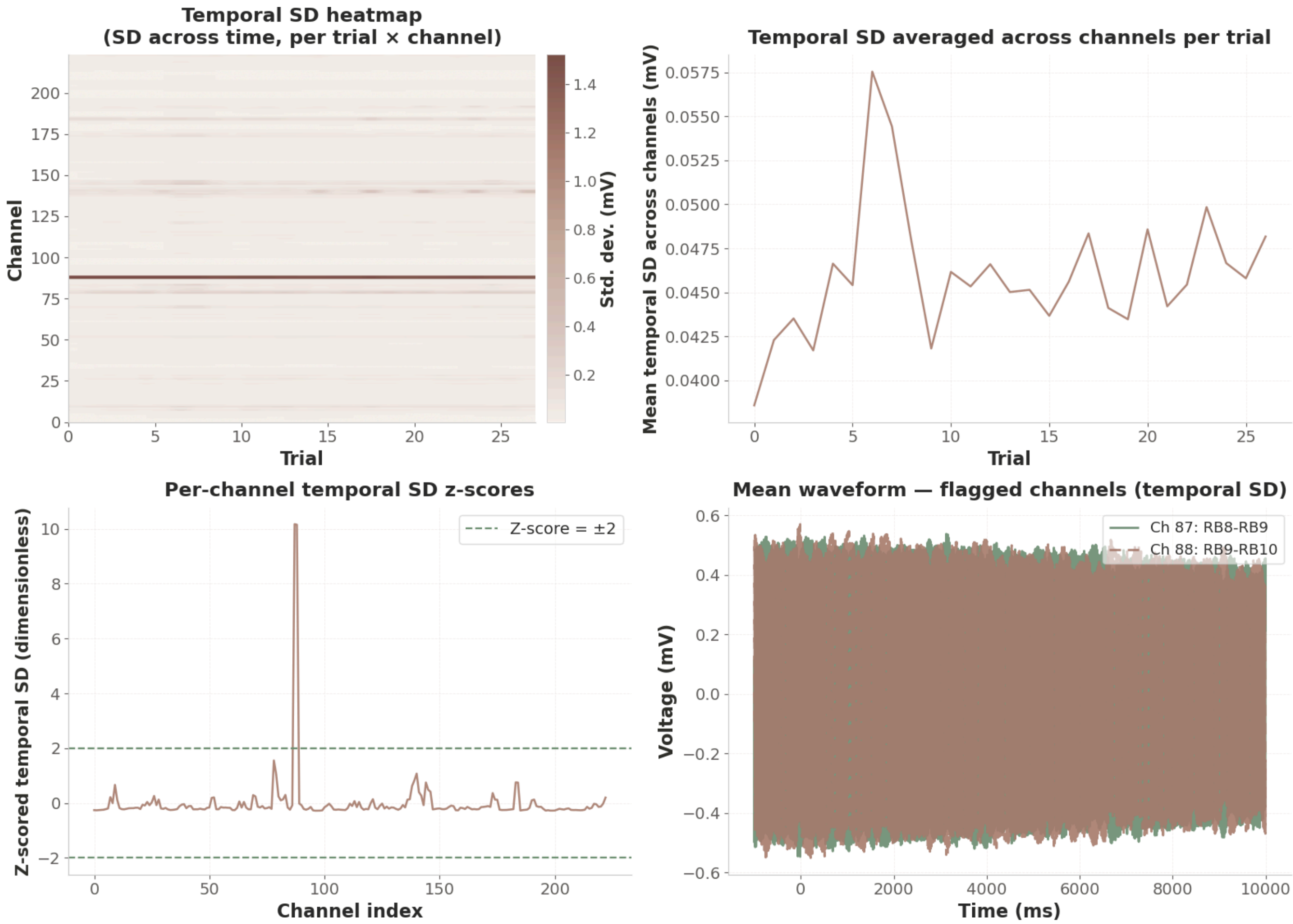
```
Total channels analyzed: 223
Channels with high variability (|z-score| > 2): 2

Potential problematic channels:
  Channel 87: RB8-RB9 (z-score: 10.17)
  Channel 88: RB9-RB10 (z-score: 10.15)
```



4.2 Temporal Variability

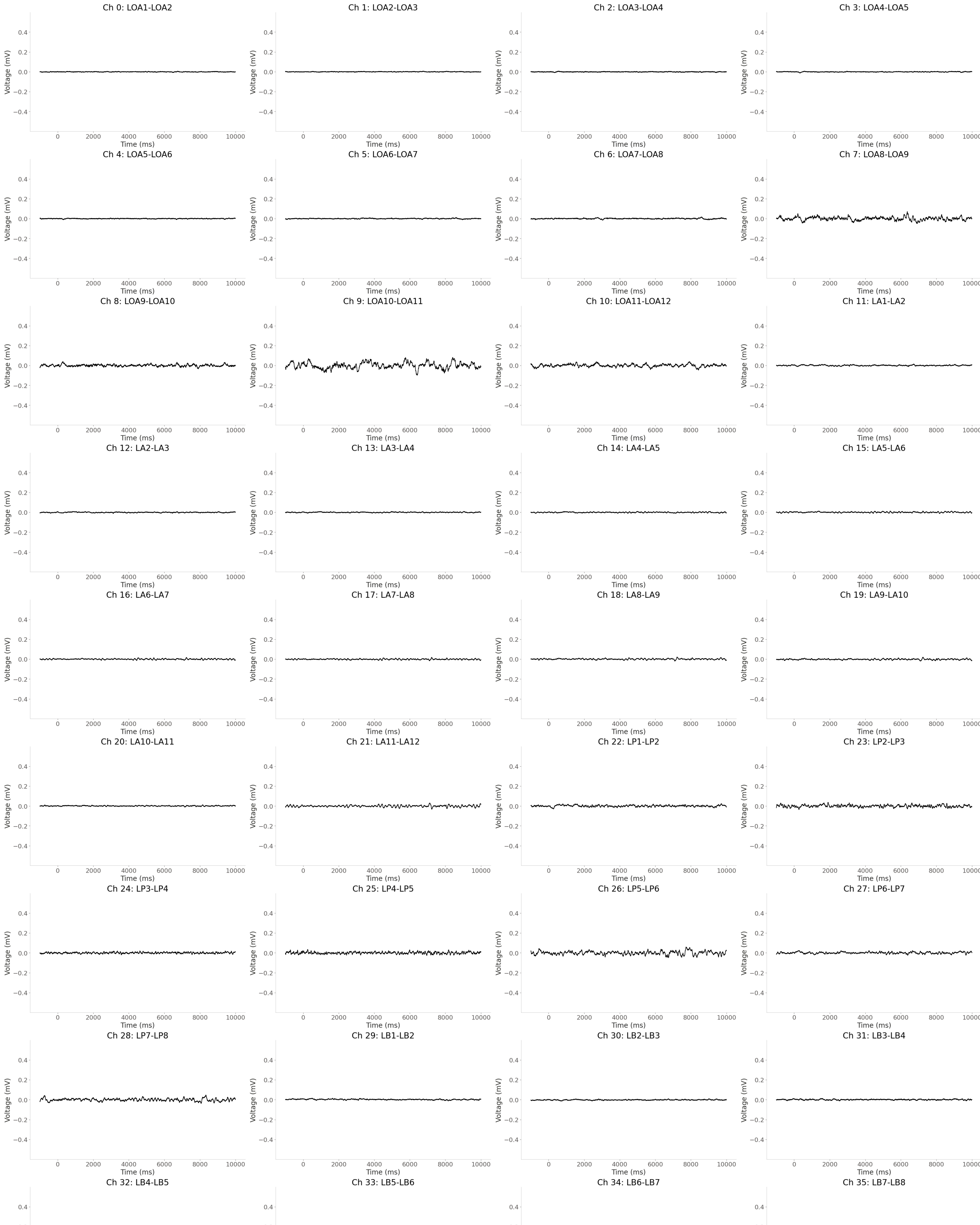
```
Total channels analyzed: 223
Channels with high temporal variability (|z-score| > 2): 2

Potential problematic channels:
  Channel 87: RB8–RB9 (z-score: 10.17)
  Channel 88: RB9–RB10 (z-score: 10.15)
```

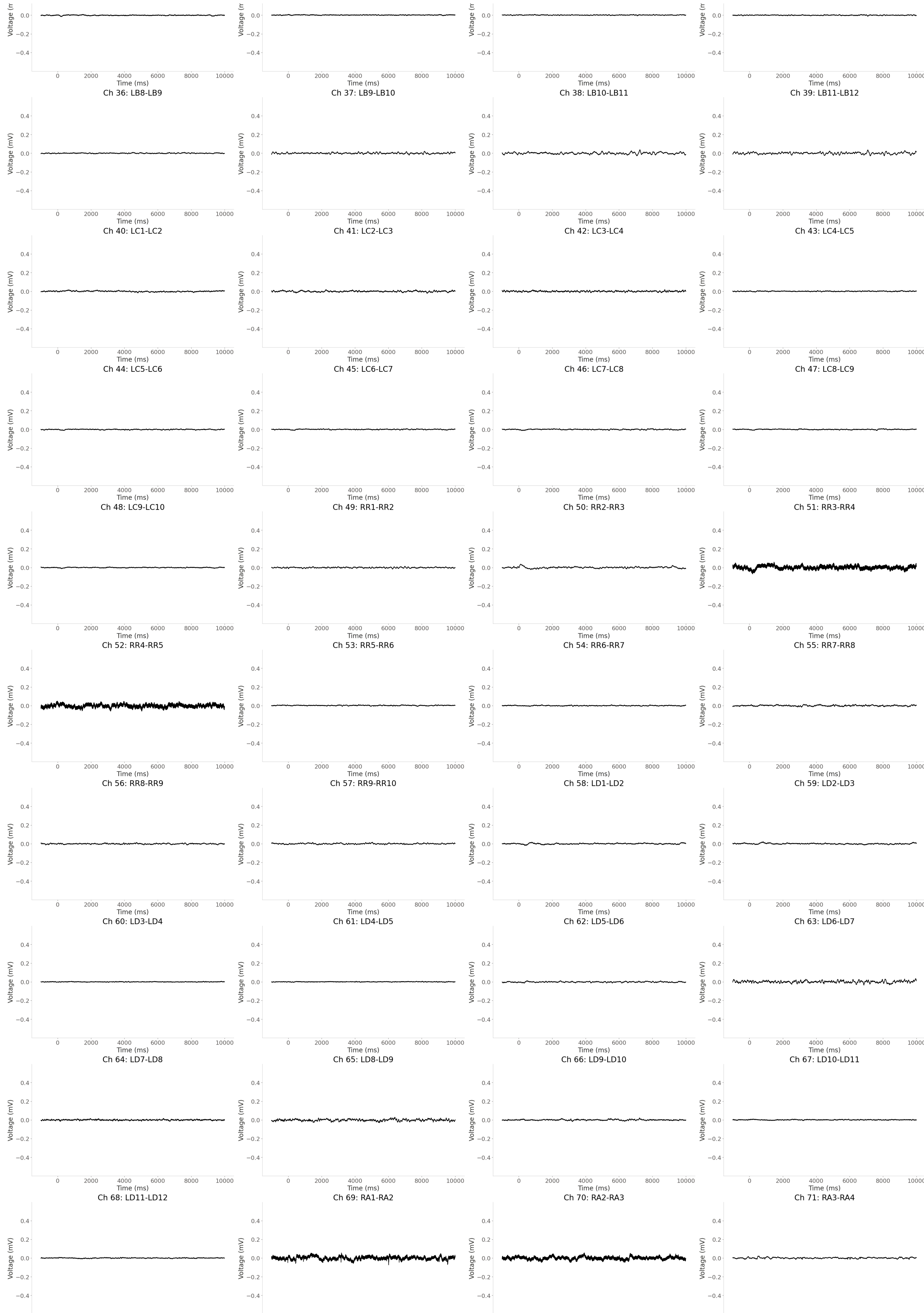
Channel quality assessment: within-trial temporal SD

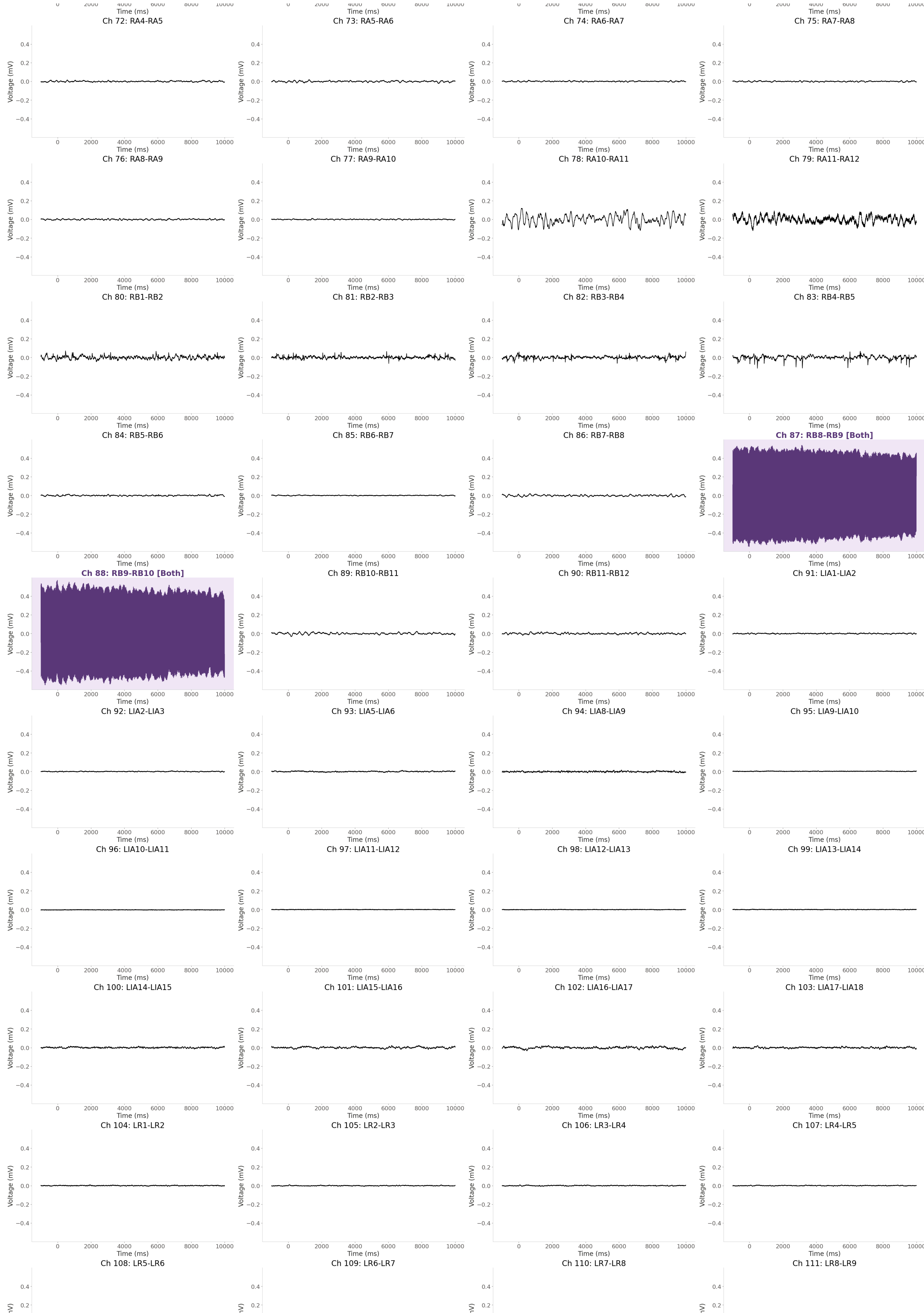


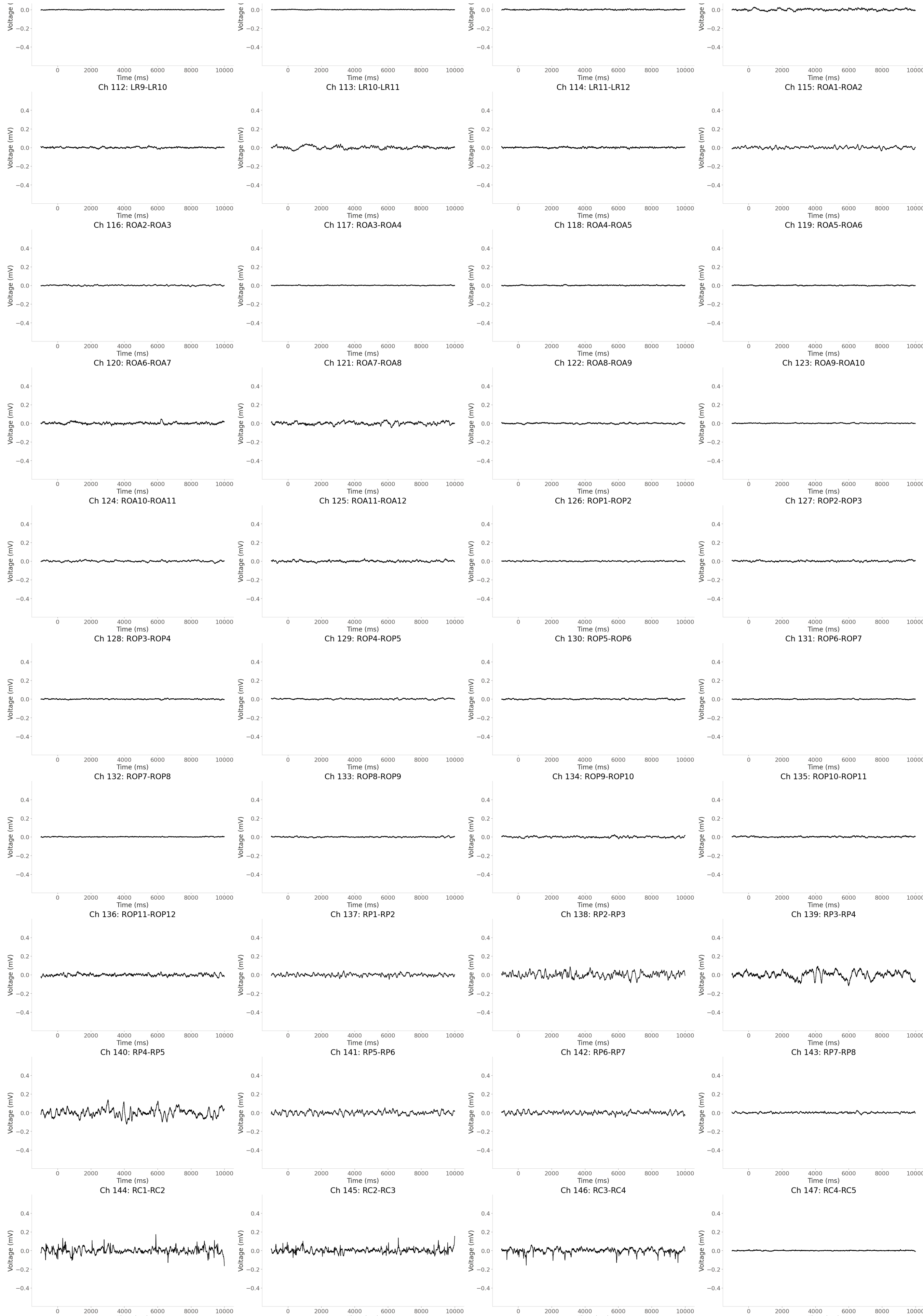

```
Channels flagged by trial variance only (green):        []
Channels flagged by temporal variance only (brown):     []
Channels flagged by BOTH (purple):                      [87, 88]
```

All channels — mean across trials
Purple = both, green = trial variance, brown = temporal variance

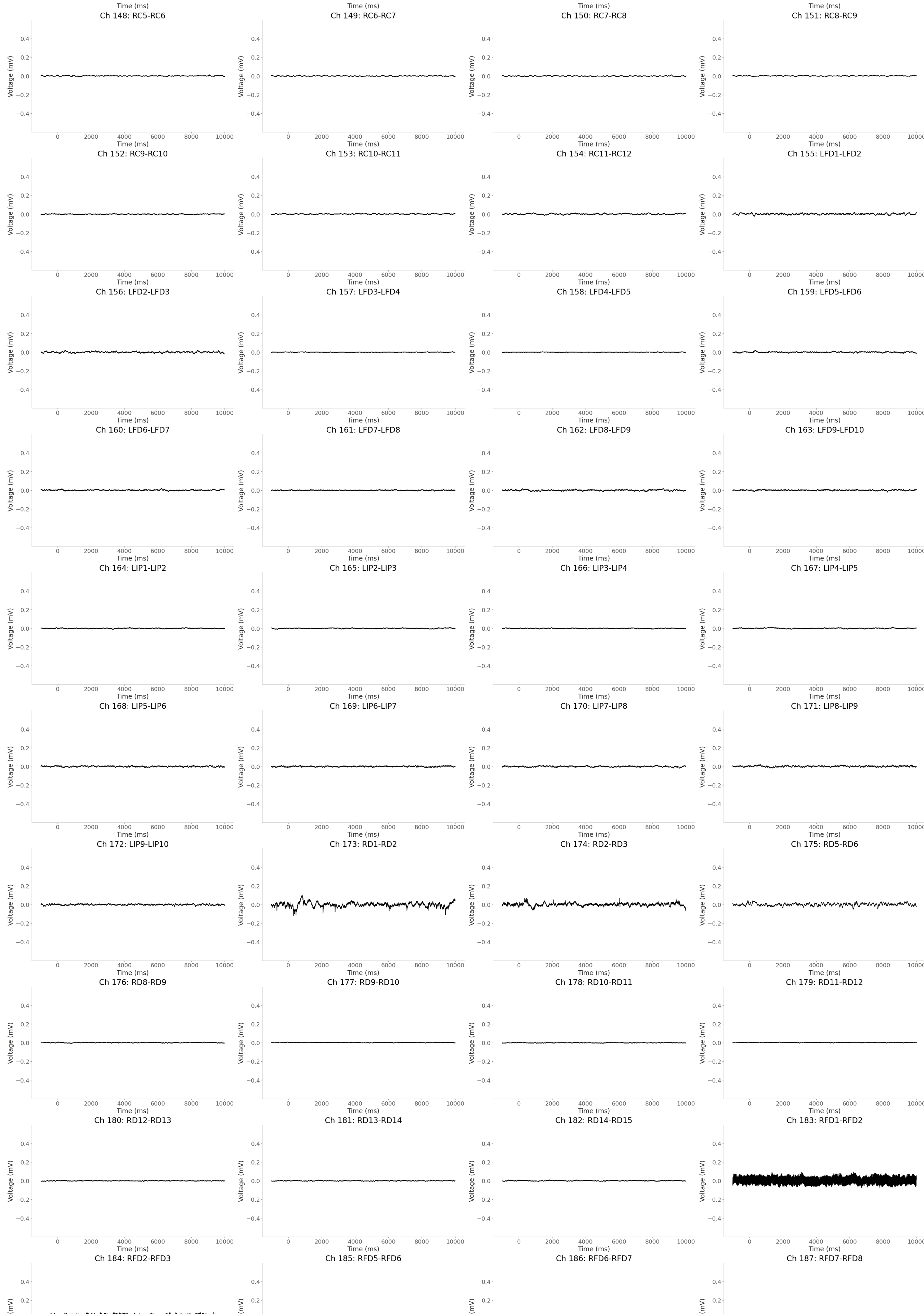


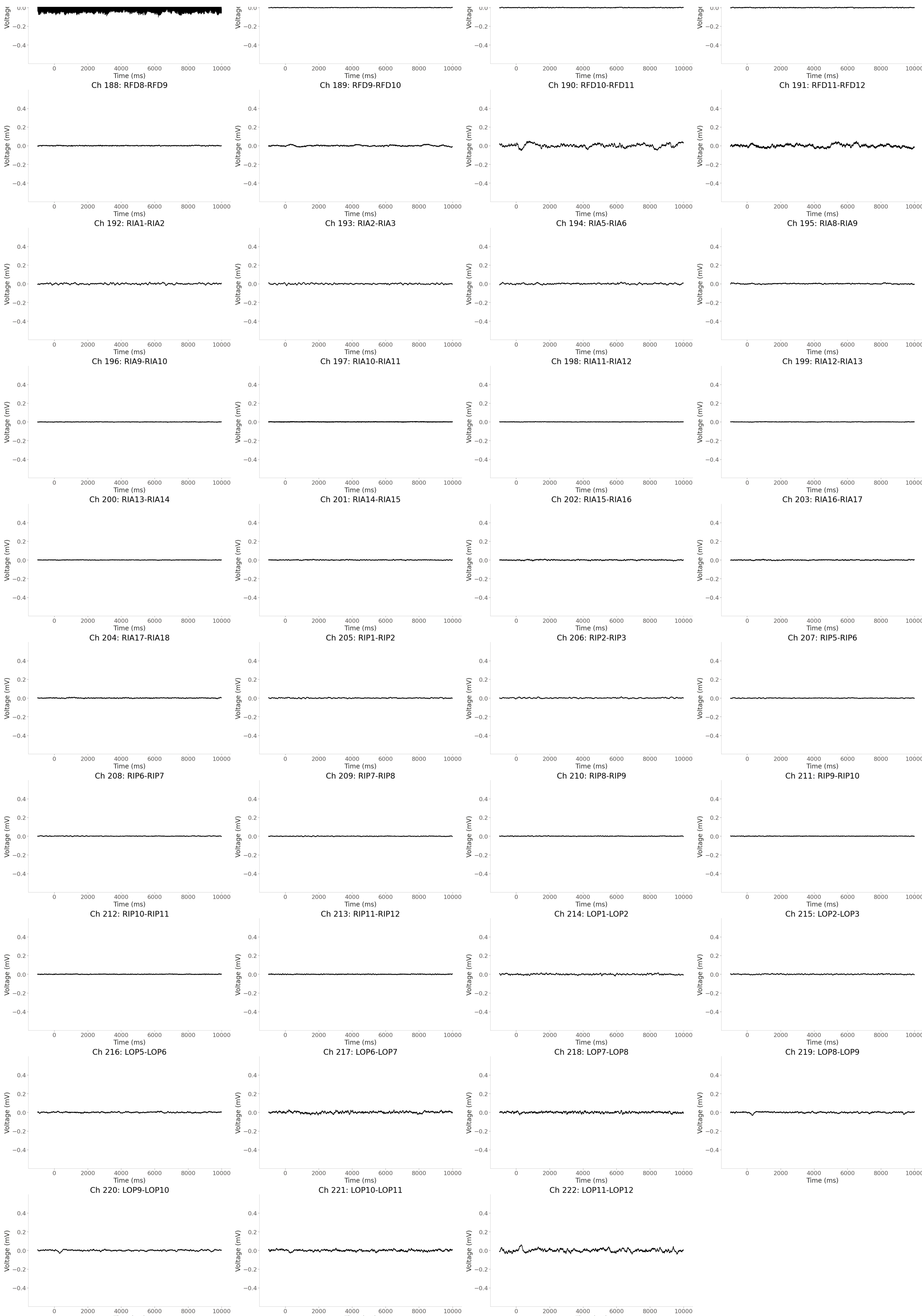

Ch 36: LB8-LB9
Ch 37: LB9-LB10
Ch 38: LB10-LB11
Ch 39: LB11-LB12
Ch 40: LC1-LC2
Ch 41: LC2-LC3
Ch 42: LC3-LC4
Ch 43: LC4-LC5
Ch 44: LC5-LC6
Ch 45: LC6-LC7
Ch 46: LC7-LC8
Ch 47: LC8-LC9
Ch 48: LC9-LC10
Ch 49: RR1-RR2
Ch 50: RR2-RR3
Ch 51: RR3-RR4
Ch 52: RR4-RR5
Ch 53: RR5-RR6
Ch 54: RR6-RR7
Ch 55: RR7-RR8
Ch 56: RR8-RR9
Ch 57: RR9-RR10
Ch 58: LD1-LD2
Ch 59: LD2-LD3
Ch 60: LD3-LD4
Ch 61: LD4-LD5
Ch 62: LD5-LD6
Ch 63: LD6-LD7
Ch 64: LD7-LD8
Ch 65: LD8-LD9
Ch 66: LD9-LD10
Ch 67: LD10-LD11
Ch 68: LD11-LD12
Ch 69: RA1-RA2
Ch 70: RA2-RA3
Ch 71: RA3-RA4
Voltage (mV)
Time (ms)


Ch 72: RA4-RA5
Ch 73: RA5-RA6
Ch 74: RA6-RA7
Ch 75: RA7-RA8
Ch 76: RA8-RA9
Ch 77: RA9-RA10
Ch 78: RA10-RA11
Ch 79: RA11-RA12
Ch 80: RB1-RB2
Ch 81: RB2-RB3
Ch 82: RB3-RB4
Ch 83: RB4-RB5
Ch 84: RB5-RB6
Ch 85: RB6-RB7
Ch 86: RB7-RB8
Ch 87: RB8-RB9 [Both]
Ch 88: RB9-RB10 [Both]
Ch 89: RB10-RB11
Ch 90: RB11-RB12
Ch 91: LIA1-LIA2
Ch 92: LIA2-LIA3
Ch 93: LIA5-LIA6
Ch 94: LIA8-LIA9
Ch 95: LIA9-LIA10
Ch 96: LIA10-LIA11
Ch 97: LIA11-LIA12
Ch 98: LIA12-LIA13
Ch 99: LIA13-LIA14
Ch 100: LIA14-LIA15
Ch 101: LIA15-LIA16
Ch 102: LIA16-LIA17
Ch 103: LIA17-LIA18
Ch 104: LR1-LR2
Ch 105: LR2-LR3
Ch 106: LR3-LR4
Ch 107: LR4-LR5
Ch 108: LR5-LR6
Ch 109: LR6-LR7
Ch 110: LR7-LR8
Ch 111: LR8-LR9
Voltage (mV)
Time (ms)



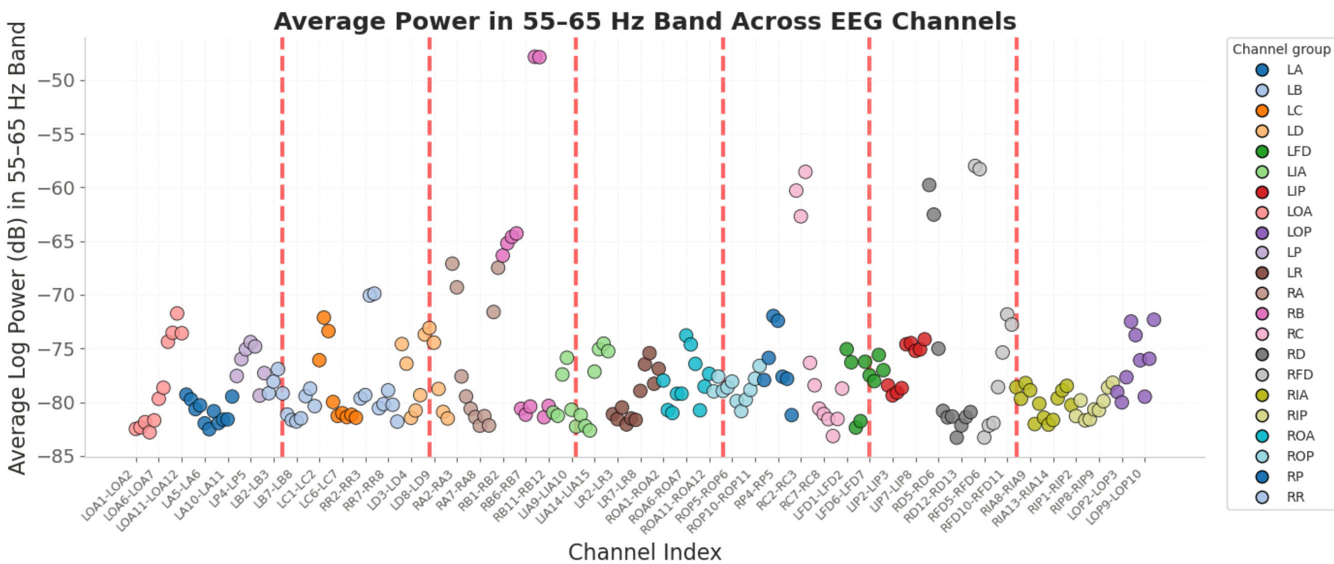
Ch 112: LR9-LR10
Ch 113: LR10-LR11
Ch 114: LR11-LR12
Ch 115: ROA1-ROA2
Ch 116: ROA2-ROA3
Ch 117: ROA3-ROA4
Ch 118: ROA4-ROA5
Ch 119: ROA5-ROA6
Ch 120: ROA6-ROA7
Ch 121: ROA7-ROA8
Ch 122: ROA8-ROA9
Ch 123: ROA9-ROA10
Ch 124: ROA10-ROA11
Ch 125: ROA11-ROA12
Ch 126: ROP1-ROP2
Ch 127: ROP2-ROP3
Ch 128: ROP3-ROP4
Ch 129: ROP4-ROP5
Ch 130: ROP5-ROP6
Ch 131: ROP6-ROP7
Ch 132: ROP7-ROP8
Ch 133: ROP8-ROP9
Ch 134: ROP9-ROP10
Ch 135: ROP10-ROP11
Ch 136: ROP11-ROP12
Ch 137: RP1-RP2
Ch 138: RP2-RP3
Ch 139: RP3-RP4
Ch 140: RP4-RP5
Ch 141: RP5-RP6
Ch 142: RP6-RP7
Ch 143: RP7-RP8
Ch 144: RC1-RC2
Ch 145: RC2-RC3
Ch 146: RC3-RC4
Ch 147: RC4-RC5
Voltage (mV)
Time (ms)


Time (ms)
Voltage (mV)
Ch 148: RC5-RC6
Ch 149: RC6-RC7
Ch 150: RC7-RC8
Ch 151: RC8-RC9
Ch 152: RC9-RC10
Ch 153: RC10-RC11
Ch 154: RC11-RC12
Ch 155: LFD1-LFD2
Ch 156: LFD2-LFD3
Ch 157: LFD3-LFD4
Ch 158: LFD4-LFD5
Ch 159: LFD5-LFD6
Ch 160: LFD6-LFD7
Ch 161: LFD7-LFD8
Ch 162: LFD8-LFD9
Ch 163: LFD9-LFD10
Ch 164: LIP1-LIP2
Ch 165: LIP2-LIP3
Ch 166: LIP3-LIP4
Ch 167: LIP4-LIP5
Ch 168: LIP5-LIP6
Ch 169: LIP6-LIP7
Ch 170: LIP7-LIP8
Ch 171: LIP8-LIP9
Ch 172: LIP9-LIP10
Ch 173: RD1-RD2
Ch 174: RD2-RD3
Ch 175: RD5-RD6
Ch 176: RD8-RD9
Ch 177: RD9-RD10
Ch 178: RD10-RD11
Ch 179: RD11-RD12
Ch 180: RD12-RD13
Ch 181: RD13-RD14
Ch 182: RD14-RD15
Ch 183: RFD1-RFD2
Ch 184: RFD2-RFD3
Ch 185: RFD5-RFD6
Ch 186: RFD6-RFD7
Ch 187: RFD7-RFD8



Ch 188: RFD8-RFD9
Ch 189: RFD9-RFD10
Ch 190: RFD10-RFD11
Ch 191: RFD11-RFD12
Ch 192: RIA1-RIA2
Ch 193: RIA2-RIA3
Ch 194: RIA5-RIA6
Ch 195: RIA8-RIA9
Ch 196: RIA9-RIA10
Ch 197: RIA10-RIA11
Ch 198: RIA11-RIA12
Ch 199: RIA12-RIA13
Ch 200: RIA13-RIA14
Ch 201: RIA14-RIA15
Ch 202: RIA15-RIA16
Ch 203: RIA16-RIA17
Ch 204: RIA17-RIA18
Ch 205: RIP1-RIP2
Ch 206: RIP2-RIP3
Ch 207: RIP5-RIP6
Ch 208: RIP6-RIP7
Ch 209: RIP7-RIP8
Ch 210: RIP8-RIP9
Ch 211: RIP9-RIP10
Ch 212: RIP10-RIP11
Ch 213: RIP11-RIP12
Ch 214: LOP1-LOP2
Ch 215: LOP2-LOP3
Ch 216: LOP5-LOP6
Ch 217: LOP6-LOP7
Ch 218: LOP7-LOP8
Ch 219: LOP8-LOP9
Ch 220: LOP9-LOP10
Ch 221: LOP10-LOP11
Ch 222: LOP11-LOP12
Voltage (mV)
Time (ms)

4.4 Line Noise Detection




```
55–65 Hz Band Power Summary:
Mean: -77.5016 dB
Std Dev: 5.5940 dB
Lowest band power:  -83.3027 dB | channel RD11-RD12
Highest band power: -47.8416 dB | channel RB8-RB9
Higher power in this band may indicate greater 60 Hz line-noise contamination.
```

```
================================================================================
Channels with low or high 55–65 Hz band power
================================================================================

Channels with Z-score < -2:
  No channels found with Z-score < -2

Channels with Z-score > 2:
  RB8-RB9          | Power: -47.8416 dB | Z-score: +5.2895
  RB9-RB10         | Power: -47.8690 dB | Z-score: +5.2846
  RFD1-RFD2        | Power: -58.0013 dB | Z-score: +3.4778
  RFD2-RFD3        | Power: -58.2965 dB | Z-score: +3.4251
  RC3-RC4          | Power: -58.5533 dB | Z-score: +3.3793
  RD1-RD2          | Power: -59.7771 dB | Z-score: +3.1611
  RC1-RC2          | Power: -60.2968 dB | Z-score: +3.0684
  RD2-RD3          | Power: -62.5284 dB | Z-score: +2.6705
  RC2-RC3          | Power: -62.6964 dB | Z-score: +2.6405
  RB4-RB5          | Power: -64.2856 dB | Z-score: +2.3571
  RB3-RB4          | Power: -64.6040 dB | Z-score: +2.3004
  RB2-RB3          | Power: -65.2032 dB | Z-score: +2.1935

================================================================================
All channel powers, sorted from highest to lowest
================================================================================
  RB8-RB9          | Power: -47.8416 dB | Z-score: +5.2895 ← Z > 2
  RB9-RB10         | Power: -47.8690 dB | Z-score: +5.2846 ← Z > 2
  RFD1-RFD2        | Power: -58.0013 dB | Z-score: +3.4778 ← Z > 2
  RFD2-RFD3        | Power: -58.2965 dB | Z-score: +3.4251 ← Z > 2
  RC3-RC4          | Power: -58.5533 dB | Z-score: +3.3793 ← Z > 2
  RD1-RD2          | Power: -59.7771 dB | Z-score: +3.1611 ← Z > 2
  RC1-RC2          | Power: -60.2968 dB | Z-score: +3.0684 ← Z > 2
  RD2-RD3          | Power: -62.5284 dB | Z-score: +2.6705 ← Z > 2
  RC2-RC3          | Power: -62.6964 dB | Z-score: +2.6405 ← Z > 2
  RB4-RB5          | Power: -64.2856 dB | Z-score: +2.3571 ← Z > 2
  RB3-RB4          | Power: -64.6040 dB | Z-score: +2.3004 ← Z > 2
  RB2-RB3          | Power: -65.2032 dB | Z-score: +2.1935 ← Z > 2
  RB1-RB2          | Power: -66.3468 dB | Z-score: +1.9896
  RA1-RA2          | Power: -67.1031 dB | Z-score: +1.8547
  RA11-RA12        | Power: -67.4815 dB | Z-score: +1.7872
  RA2-RA3          | Power: -69.3056 dB | Z-score: +1.4619
  RR4-RR5          | Power: -69.8916 dB | Z-score: +1.3574
  RR3-RR4          | Power: -70.0677 dB | Z-score: +1.3260
  RA10-RA11        | Power: -71.6012 dB | Z-score: +1.0526
  LOA10-LOA11      | Power: -71.7337 dB | Z-score: +1.0289
  RFD10-RFD11      | Power: -71.8263 dB | Z-score: +1.0124
  RP3-RP4          | Power: -71.9688 dB | Z-score: +0.9870
  LC2-LC3          | Power: -72.1182 dB | Z-score: +0.9604
  LOP11-LOP12      | Power: -72.3222 dB | Z-score: +0.9240
  RP4-RP5          | Power: -72.4213 dB | Z-score: +0.9063
  LOP6-LOP7        | Power: -72.4871 dB | Z-score: +0.8946
  RFD11-RFD12      | Power: -72.7715 dB | Z-score: +0.8439
  LD7-LD8          | Power: -73.0877 dB | Z-score: +0.7875
  LC3-LC4          | Power: -73.3782 dB | Z-score: +0.7357
  LOA9-LOA10       | Power: -73.5407 dB | Z-score: +0.7067
  LOA11-LOA12      | Power: -73.5762 dB | Z-score: +0.7004
```

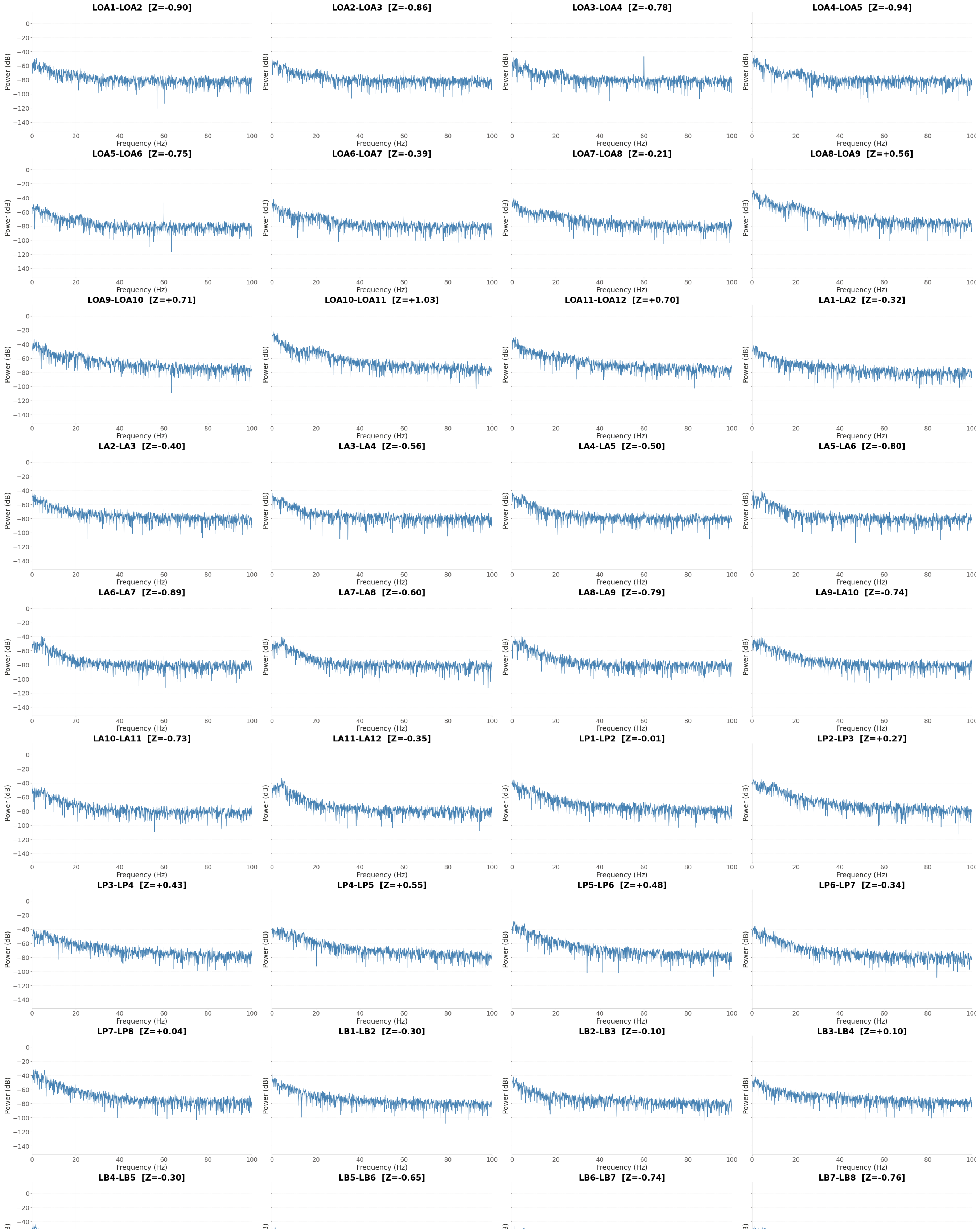
```
LD6-LD7          | Power: -73.6960 dB | Z-score: +0.6790
LOP7-LOP8        | Power: -73.7571 dB | Z-score: +0.6681
ROA6-ROA7        | Power: -73.7887 dB | Z-score: +0.6625
LIP9-LIP10       | Power: -74.1453 dB | Z-score: +0.5989
LOA8-LOA9        | Power: -74.3722 dB | Z-score: +0.5584
LP4-LP5          | Power: -74.4057 dB | Z-score: +0.5525
LD8-LD9          | Power: -74.4563 dB | Z-score: +0.5434
LIP6-LIP7        | Power: -74.4989 dB | Z-score: +0.5358
LIA16-LIA17      | Power: -74.5453 dB | Z-score: +0.5276
LD1-LD2          | Power: -74.5888 dB | Z-score: +0.5198
LIP5-LIP6        | Power: -74.6093 dB | Z-score: +0.5161
ROA7-ROA8        | Power: -74.6256 dB | Z-score: +0.5132
LP5-LP6          | Power: -74.8111 dB | Z-score: +0.4802
RD5-RD6          | Power: -75.0136 dB | Z-score: +0.4441
LIA15-LIA16      | Power: -75.0621 dB | Z-score: +0.4354
LFD1-LFD2        | Power: -75.0629 dB | Z-score: +0.4353
LP3-LP4          | Power: -75.0762 dB | Z-score: +0.4329
LIP8-LIP9        | Power: -75.0830 dB | Z-score: +0.4317
LIP7-LIP8        | Power: -75.2252 dB | Z-score: +0.4063
LIA17-LIA18      | Power: -75.2538 dB | Z-score: +0.4012
RFD9-RFD10       | Power: -75.3710 dB | Z-score: +0.3803
LR9-LR10         | Power: -75.4357 dB | Z-score: +0.3688
LFD8-LFD9        | Power: -75.6012 dB | Z-score: +0.3393
LIA8-LIA9        | Power: -75.8624 dB | Z-score: +0.2927
RP2-RP3          | Power: -75.8776 dB | Z-score: +0.2900
LOP10-LOP11      | Power: -75.9608 dB | Z-score: +0.2751
LP2-LP3          | Power: -75.9899 dB | Z-score: +0.2700
LC1-LC2          | Power: -76.0887 dB | Z-score: +0.2523
LOP8-LOP9        | Power: -76.1006 dB | Z-score: +0.2502
LFD5-LFD6        | Power: -76.2314 dB | Z-score: +0.2269
LFD2-LFD3        | Power: -76.2842 dB | Z-score: +0.2175
RC4-RC5          | Power: -76.3378 dB | Z-score: +0.2079
LD2-LD3          | Power: -76.4117 dB | Z-score: +0.1947
ROA8-ROA9        | Power: -76.4300 dB | Z-score: +0.1915
LR8-LR9          | Power: -76.4649 dB | Z-score: +0.1852
ROP11-ROP12      | Power: -76.6238 dB | Z-score: +0.1569
LR11-LR12        | Power: -76.8898 dB | Z-score: +0.1095
LB3-LB4          | Power: -76.9209 dB | Z-score: +0.1039
LFD9-LFD10       | Power: -77.0218 dB | Z-score: +0.0859
LIA14-LIA15      | Power: -77.1564 dB | Z-score: +0.0619
LP7-LP8          | Power: -77.2902 dB | Z-score: +0.0381
ROA11-ROA12      | Power: -77.3602 dB | Z-score: +0.0256
LIA5-LIA6        | Power: -77.4203 dB | Z-score: +0.0149
LFD6-LFD7        | Power: -77.4994 dB | Z-score: +0.0008
LP1-LP2          | Power: -77.5589 dB | Z-score: -0.0098
RA3-RA4          | Power: -77.6015 dB | Z-score: -0.0174
ROP2-ROP3        | Power: -77.6155 dB | Z-score: -0.0199
RP5-RP6          | Power: -77.6269 dB | Z-score: -0.0220
LOP5-LOP6        | Power: -77.6762 dB | Z-score: -0.0307
ROP10-ROP11      | Power: -77.8196 dB | Z-score: -0.0563
RP6-RP7          | Power: -77.8245 dB | Z-score: -0.0572
RP1-RP2          | Power: -77.9367 dB | Z-score: -0.0772
ROA1-ROA2        | Power: -77.9798 dB | Z-score: -0.0849
LFD7-LFD8        | Power: -78.0468 dB | Z-score: -0.0968
ROP5-ROP6        | Power: -78.0786 dB | Z-score: -0.1025
LB2-LB3          | Power: -78.0842 dB | Z-score: -0.1035
```

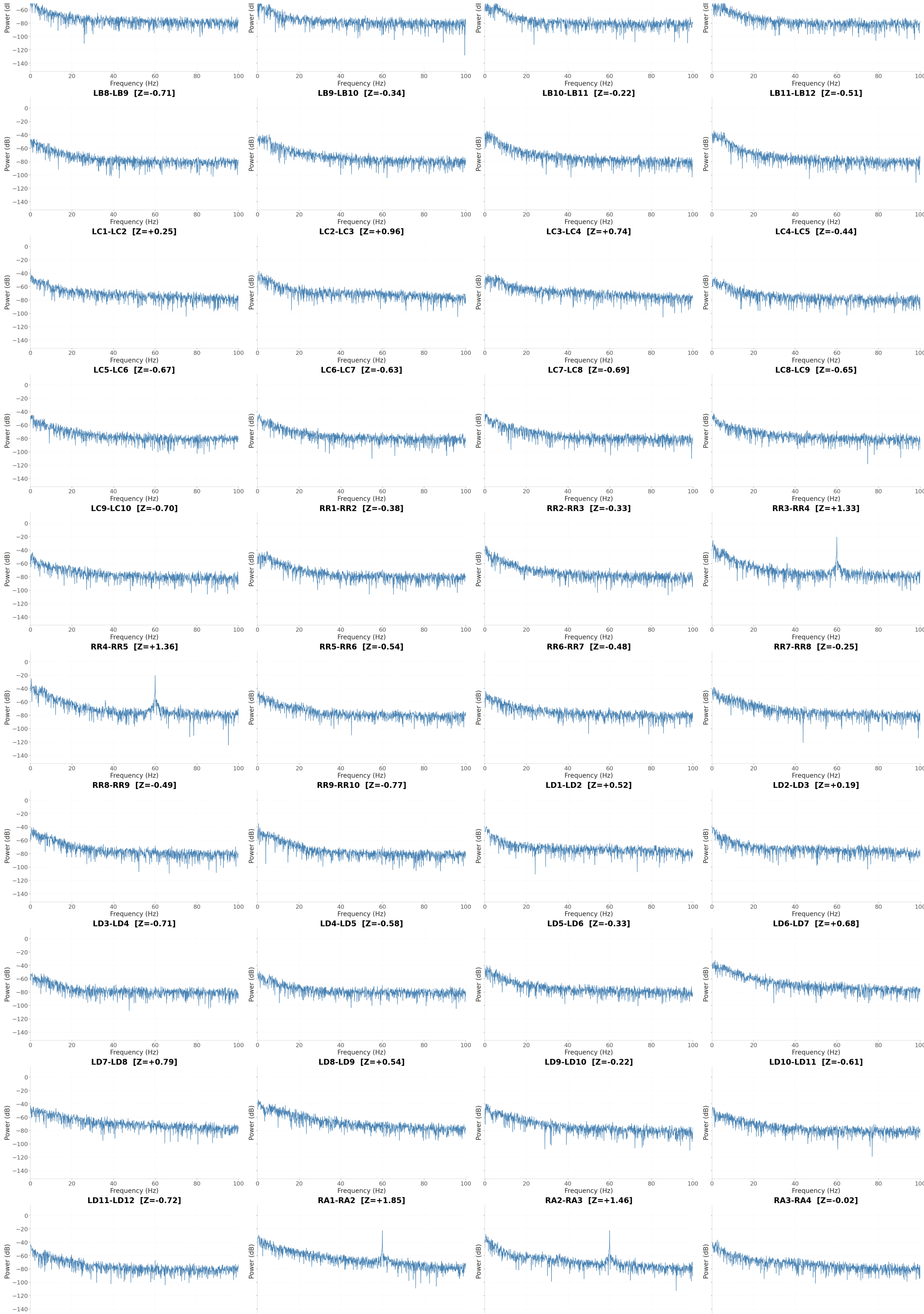
```
RIP11-RIP12     | Power: -78.1935 dB | Z-score: -0.1230
RIA5-RIA6       | Power: -78.2212 dB | Z-score: -0.1279
LR10-LR11       | Power: -78.2857 dB | Z-score: -0.1394
RC5-RC6         | Power: -78.4269 dB | Z-score: -0.1646
LIP1-LIP2       | Power: -78.4320 dB | Z-score: -0.1655
RIA16-RIA17     | Power: -78.4805 dB | Z-score: -0.1742
ROA10-ROA11     | Power: -78.5412 dB | Z-score: -0.1850
RFD8-RFD9       | Power: -78.5960 dB | Z-score: -0.1948
RIA1-RIA2       | Power: -78.6097 dB | Z-score: -0.1972
ROP4-ROP5       | Power: -78.6286 dB | Z-score: -0.2006
RIP10-RIP11     | Power: -78.6364 dB | Z-score: -0.2020
LOA7-LOA8       | Power: -78.6541 dB | Z-score: -0.2051
LIP4-LIP5       | Power: -78.6901 dB | Z-score: -0.2116
RC11-RC12       | Power: -78.7334 dB | Z-score: -0.2193
LB10-LB11       | Power: -78.7370 dB | Z-score: -0.2199
LD9-LD10        | Power: -78.7650 dB | Z-score: -0.2249
ROP9-ROP10      | Power: -78.8580 dB | Z-score: -0.2415
RIA8-RIA9       | Power: -78.8948 dB | Z-score: -0.2481
RIA15-RIA16     | Power: -78.8975 dB | Z-score: -0.2485
RR7-RR8         | Power: -78.8996 dB | Z-score: -0.2489
LR7-LR8         | Power: -78.9505 dB | Z-score: -0.2580
ROP3-ROP4       | Power: -78.9509 dB | Z-score: -0.2581
ROP1-ROP2       | Power: -78.9949 dB | Z-score: -0.2659
LOP1-LOP2       | Power: -79.0454 dB | Z-score: -0.2749
LIP3-LIP4       | Power: -79.0972 dB | Z-score: -0.2842
LB1-LB2         | Power: -79.1677 dB | Z-score: -0.2967
LB4-LB5         | Power: -79.1811 dB | Z-score: -0.2991
ROA5-ROA6       | Power: -79.2127 dB | Z-score: -0.3048
ROA4-ROA5       | Power: -79.2152 dB | Z-score: -0.3052
LA1-LA2         | Power: -79.2963 dB | Z-score: -0.3197
RR2-RR3         | Power: -79.3397 dB | Z-score: -0.3274
LD5-LD6         | Power: -79.3462 dB | Z-score: -0.3286
LIP2-LIP3       | Power: -79.3703 dB | Z-score: -0.3329
LP6-LP7         | Power: -79.3865 dB | Z-score: -0.3357
LB9-LB10        | Power: -79.4349 dB | Z-score: -0.3444
RA4-RA5         | Power: -79.4716 dB | Z-score: -0.3509
LA11-LA12       | Power: -79.4738 dB | Z-score: -0.3513
LOP9-LOP10      | Power: -79.4883 dB | Z-score: -0.3539
RIA14-RIA15     | Power: -79.6341 dB | Z-score: -0.3799
RR1-RR2         | Power: -79.6484 dB | Z-score: -0.3824
RIA2-RIA3       | Power: -79.6840 dB | Z-score: -0.3888
LOA6-LOA7       | Power: -79.6847 dB | Z-score: -0.3889
LA2-LA3         | Power: -79.7432 dB | Z-score: -0.3993
ROP8-ROP9       | Power: -79.7700 dB | Z-score: -0.4041
RIP2-RIP3       | Power: -79.8311 dB | Z-score: -0.4150
RIP9-RIP10      | Power: -79.9066 dB | Z-score: -0.4285
ROP6-ROP7       | Power: -79.9067 dB | Z-score: -0.4285
LC4-LC5         | Power: -79.9843 dB | Z-score: -0.4423
LOP2-LOP3       | Power: -80.0184 dB | Z-score: -0.4484
RIA10-RIA11     | Power: -80.1468 dB | Z-score: -0.4713
RR6-RR7         | Power: -80.1771 dB | Z-score: -0.4767
RR8-RR9         | Power: -80.2392 dB | Z-score: -0.4878
RIA17-RIA18     | Power: -80.2778 dB | Z-score: -0.4947
LA4-LA5         | Power: -80.2908 dB | Z-score: -0.4970
RB11-RB12       | Power: -80.3428 dB | Z-score: -0.5063
LB11-LB12       | Power: -80.3669 dB | Z-score: -0.5106
```

```
RB7-RB8          | Power: -80.4046 dB | Z-score: -0.5173
LR3-LR4          | Power: -80.4930 dB | Z-score: -0.5331
RR5-RR6          | Power: -80.5584 dB | Z-score: -0.5447
RA5-RA6          | Power: -80.6116 dB | Z-score: -0.5542
RB5-RB6          | Power: -80.6125 dB | Z-score: -0.5544
RC6-RC7          | Power: -80.6164 dB | Z-score: -0.5551
LA3-LA4          | Power: -80.6473 dB | Z-score: -0.5606
RIP7-RIP8        | Power: -80.6787 dB | Z-score: -0.5662
LIA9-LIA10       | Power: -80.7120 dB | Z-score: -0.5721
ROA2-ROA3        | Power: -80.7363 dB | Z-score: -0.5764
RIP8-RIP9        | Power: -80.7454 dB | Z-score: -0.5781
ROA9-ROA10       | Power: -80.7589 dB | Z-score: -0.5805
LD4-LD5          | Power: -80.7755 dB | Z-score: -0.5834
RD8-RD9          | Power: -80.8165 dB | Z-score: -0.5908
ROP7-ROP8        | Power: -80.8341 dB | Z-score: -0.5939
LA7-LA8          | Power: -80.8409 dB | Z-score: -0.5951
LD10-LD11        | Power: -80.9103 dB | Z-score: -0.6075
RD14-RD15        | Power: -80.9323 dB | Z-score: -0.6114
LIA1-LIA2        | Power: -80.9636 dB | Z-score: -0.6170
ROA3-ROA4        | Power: -81.0062 dB | Z-score: -0.6246
LC6-LC7          | Power: -81.0230 dB | Z-score: -0.6276
RC7-RC8          | Power: -81.1200 dB | Z-score: -0.6449
LR1-LR2          | Power: -81.1218 dB | Z-score: -0.6452
LB5-LB6          | Power: -81.1503 dB | Z-score: -0.6503
RB6-RB7          | Power: -81.1585 dB | Z-score: -0.6517
LC8-LC9          | Power: -81.1725 dB | Z-score: -0.6542
LIA11-LIA12      | Power: -81.2036 dB | Z-score: -0.6598
RP7-RP8          | Power: -81.2048 dB | Z-score: -0.6600
LIA2-LIA3        | Power: -81.2636 dB | Z-score: -0.6705
LC5-LC6          | Power: -81.2789 dB | Z-score: -0.6732
RIP1-RIP2        | Power: -81.2911 dB | Z-score: -0.6754
RA8-RA9          | Power: -81.2982 dB | Z-score: -0.6767
RD10-RD11        | Power: -81.3297 dB | Z-score: -0.6823
RD13-RD14        | Power: -81.3843 dB | Z-score: -0.6920
RA6-RA7          | Power: -81.3879 dB | Z-score: -0.6926
LC7-LC8          | Power: -81.3913 dB | Z-score: -0.6933
RB10-RB11        | Power: -81.4017 dB | Z-score: -0.6951
RD9-RD10         | Power: -81.4131 dB | Z-score: -0.6971
RIA11-RIA12      | Power: -81.4419 dB | Z-score: -0.7023
LC9-LC10         | Power: -81.4419 dB | Z-score: -0.7023
LD3-LD4          | Power: -81.4617 dB | Z-score: -0.7058
LB8-LB9          | Power: -81.5058 dB | Z-score: -0.7137
LD11-LD12        | Power: -81.5161 dB | Z-score: -0.7155
LR5-LR6          | Power: -81.5180 dB | Z-score: -0.7158
RC10-RC11        | Power: -81.5703 dB | Z-score: -0.7252
RC8-RC9          | Power: -81.5771 dB | Z-score: -0.7264
LR2-LR3          | Power: -81.5854 dB | Z-score: -0.7279
RIP6-RIP7        | Power: -81.5977 dB | Z-score: -0.7301
LA10-LA11        | Power: -81.6010 dB | Z-score: -0.7307
LA9-LA10         | Power: -81.6331 dB | Z-score: -0.7364
LR6-LR7          | Power: -81.6479 dB | Z-score: -0.7390
LB6-LB7          | Power: -81.6552 dB | Z-score: -0.7403
RIA13-RIA14      | Power: -81.6564 dB | Z-score: -0.7405
RIP5-RIP6        | Power: -81.6947 dB | Z-score: -0.7473
LOA5-LOA6        | Power: -81.7075 dB | Z-score: -0.7496
LFD4-LFD5        | Power: -81.7825 dB | Z-score: -0.7630
```

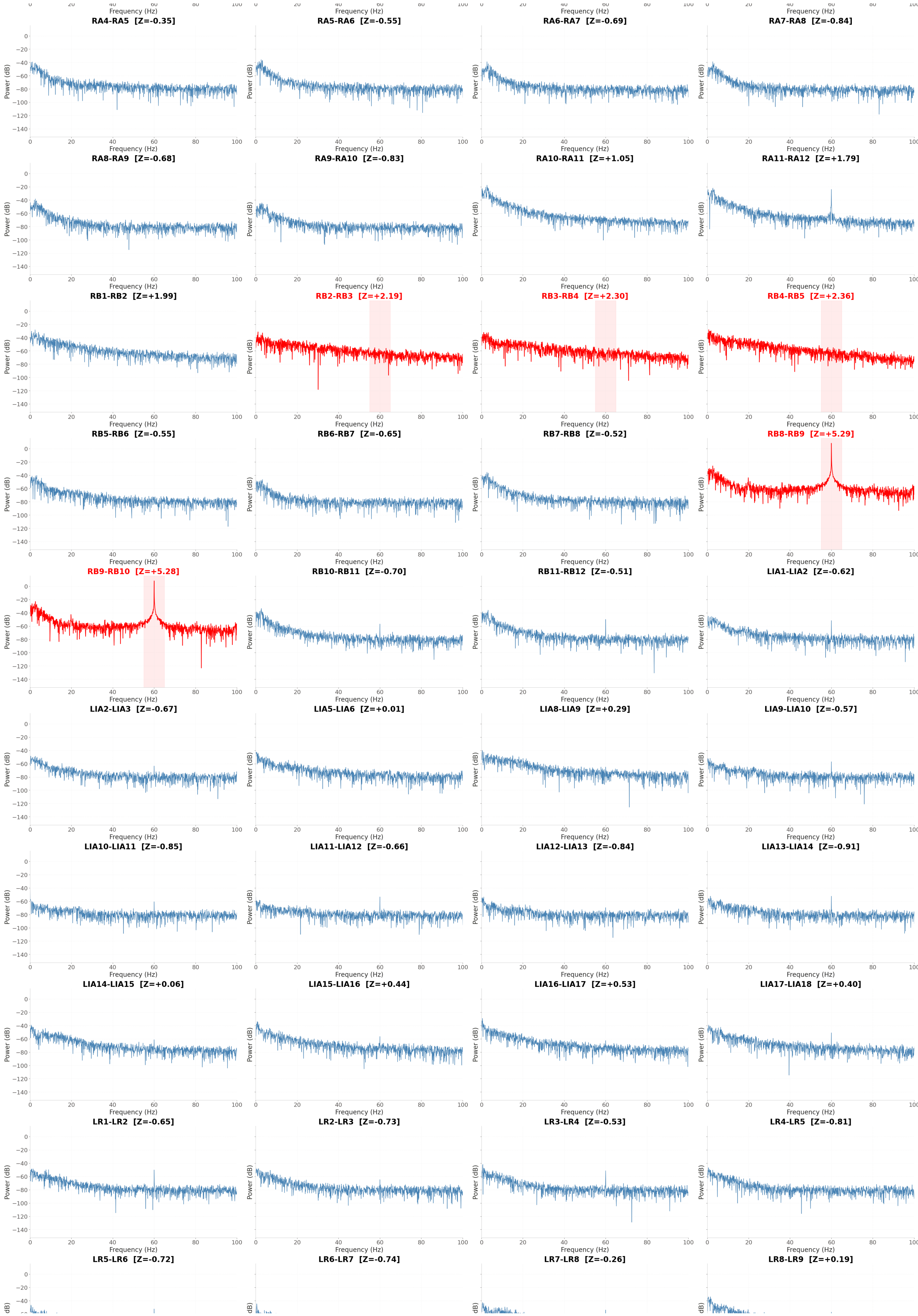
```
LB7-LB8          | Power: -81.7926 dB | Z-score: -0.7648
RR9-RR10         | Power: -81.8092 dB | Z-score: -0.7678
LOA3-LOA4        | Power: -81.8606 dB | Z-score: -0.7769
LA8-LA9          | Power: -81.9492 dB | Z-score: -0.7927
RFD7-RFD8        | Power: -81.9543 dB | Z-score: -0.7937
LA5-LA6          | Power: -81.9682 dB | Z-score: -0.7961
RIA9-RIA10       | Power: -82.0392 dB | Z-score: -0.8088
LR4-LR5          | Power: -82.0711 dB | Z-score: -0.8145
RIA12-RIA13      | Power: -82.0928 dB | Z-score: -0.8183
RD12-RD13        | Power: -82.1661 dB | Z-score: -0.8314
RA9-RA10         | Power: -82.1730 dB | Z-score: -0.8327
RFD6-RFD7        | Power: -82.1767 dB | Z-score: -0.8333
RA7-RA8          | Power: -82.1893 dB | Z-score: -0.8356
LIA12-LIA13      | Power: -82.2377 dB | Z-score: -0.8442
LIA10-LIA11      | Power: -82.2716 dB | Z-score: -0.8502
LOA2-LOA3        | Power: -82.3463 dB | Z-score: -0.8636
LFD3-LFD4        | Power: -82.3703 dB | Z-score: -0.8678
LA6-LA7          | Power: -82.5223 dB | Z-score: -0.8949
LOA1-LOA2        | Power: -82.5247 dB | Z-score: -0.8954
LIA13-LIA14      | Power: -82.6310 dB | Z-score: -0.9143
LOA4-LOA5        | Power: -82.8002 dB | Z-score: -0.9445
RC9-RC10         | Power: -83.1519 dB | Z-score: -1.0072
RFD5-RFD6        | Power: -83.2910 dB | Z-score: -1.0320
RD11-RD12        | Power: -83.3068 dB | Z-score: -1.0348
```

Line-noise power by channel
Red = 55–65 Hz band-power Z-score > 2

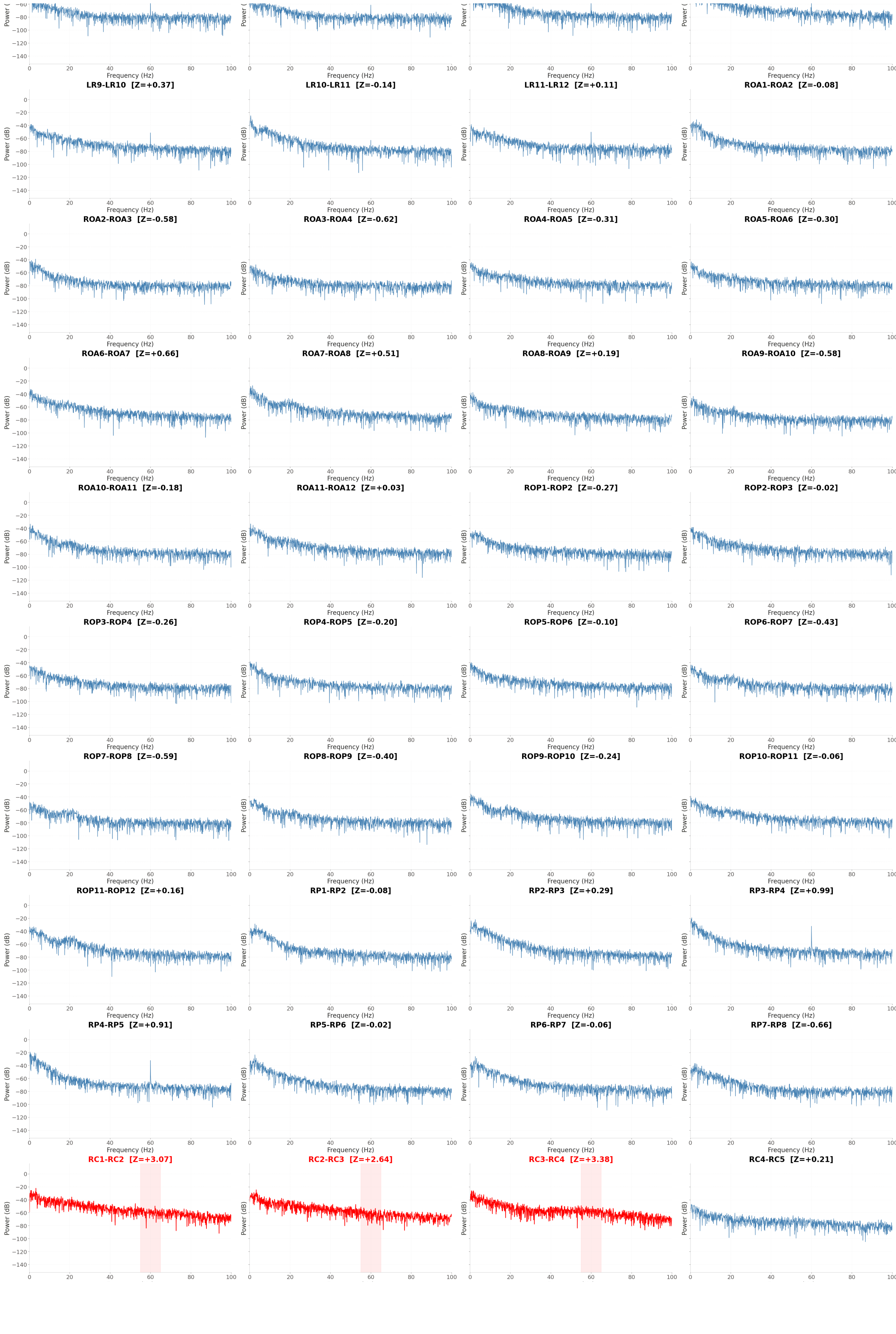




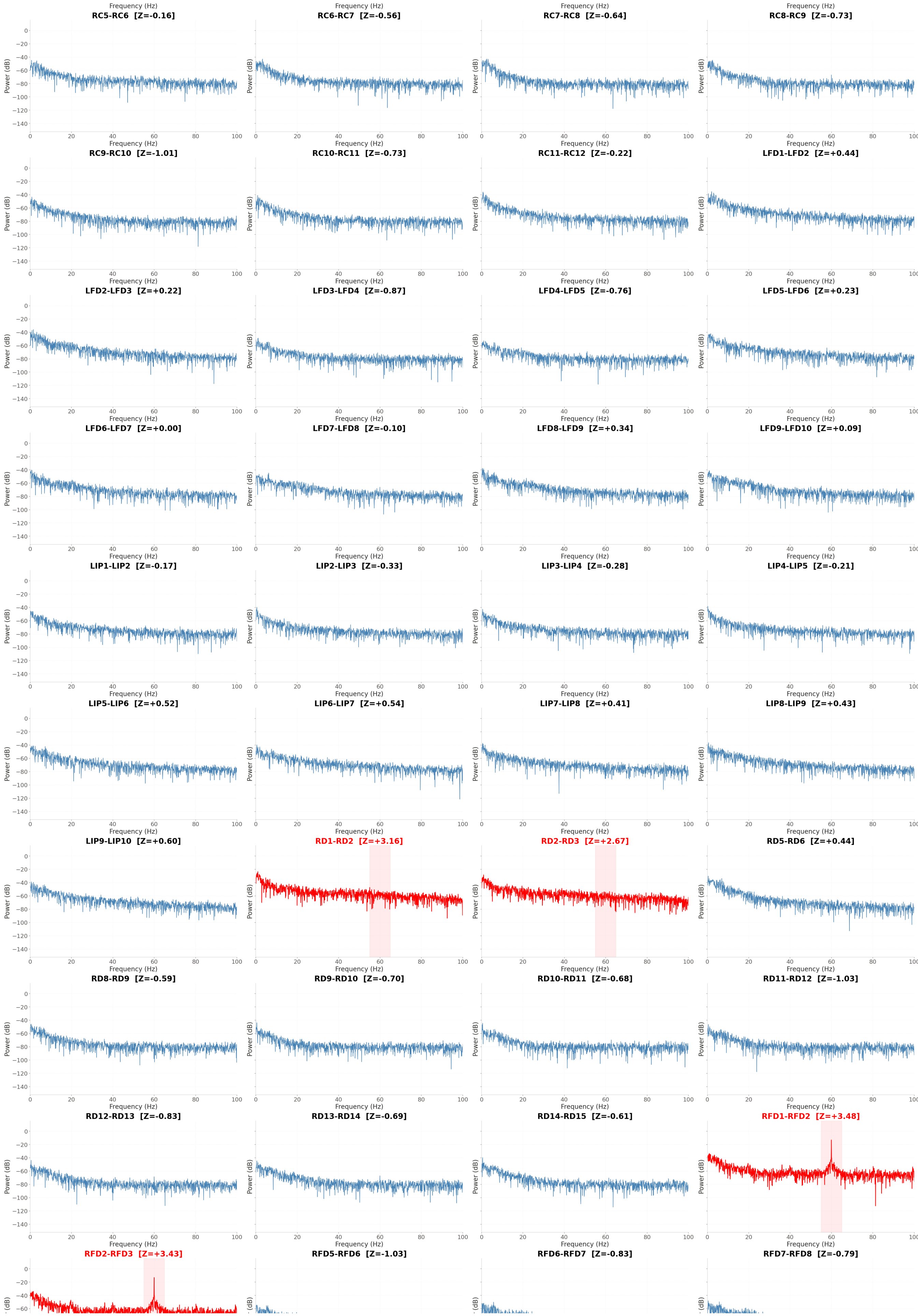
Power (dB)
Frequency (Hz)
LB8-LB9 [Z=-0.71]
LB9-LB10 [Z=-0.34]
LB10-LB11 [Z=-0.22]
LB11-LB12 [Z=-0.51]
LC1-LC2 [Z=+0.25]
LC2-LC3 [Z=+0.96]
LC3-LC4 [Z=+0.74]
LC4-LC5 [Z=-0.44]
LC5-LC6 [Z=-0.67]
LC6-LC7 [Z=-0.63]
LC7-LC8 [Z=-0.69]
LC8-LC9 [Z=-0.65]
LC9-LC10 [Z=-0.70]
RR1-RR2 [Z=-0.38]
RR2-RR3 [Z=-0.33]
RR3-RR4 [Z=+1.33]
RR4-RR5 [Z=+1.36]
RR5-RR6 [Z=-0.54]
RR6-RR7 [Z=-0.48]
RR7-RR8 [Z=-0.25]
RR8-RR9 [Z=-0.49]
RR9-RR10 [Z=-0.77]
LD1-LD2 [Z=+0.52]
LD2-LD3 [Z=+0.19]
LD3-LD4 [Z=-0.71]
LD4-LD5 [Z=-0.58]
LD5-LD6 [Z=-0.33]
LD6-LD7 [Z=+0.68]
LD7-LD8 [Z=+0.79]
LD8-LD9 [Z=+0.54]
LD9-LD10 [Z=-0.22]
LD10-LD11 [Z=-0.61]
LD11-LD12 [Z=-0.72]
RA1-RA2 [Z=+1.85]
RA2-RA3 [Z=+1.46]
RA3-RA4 [Z=-0.02]

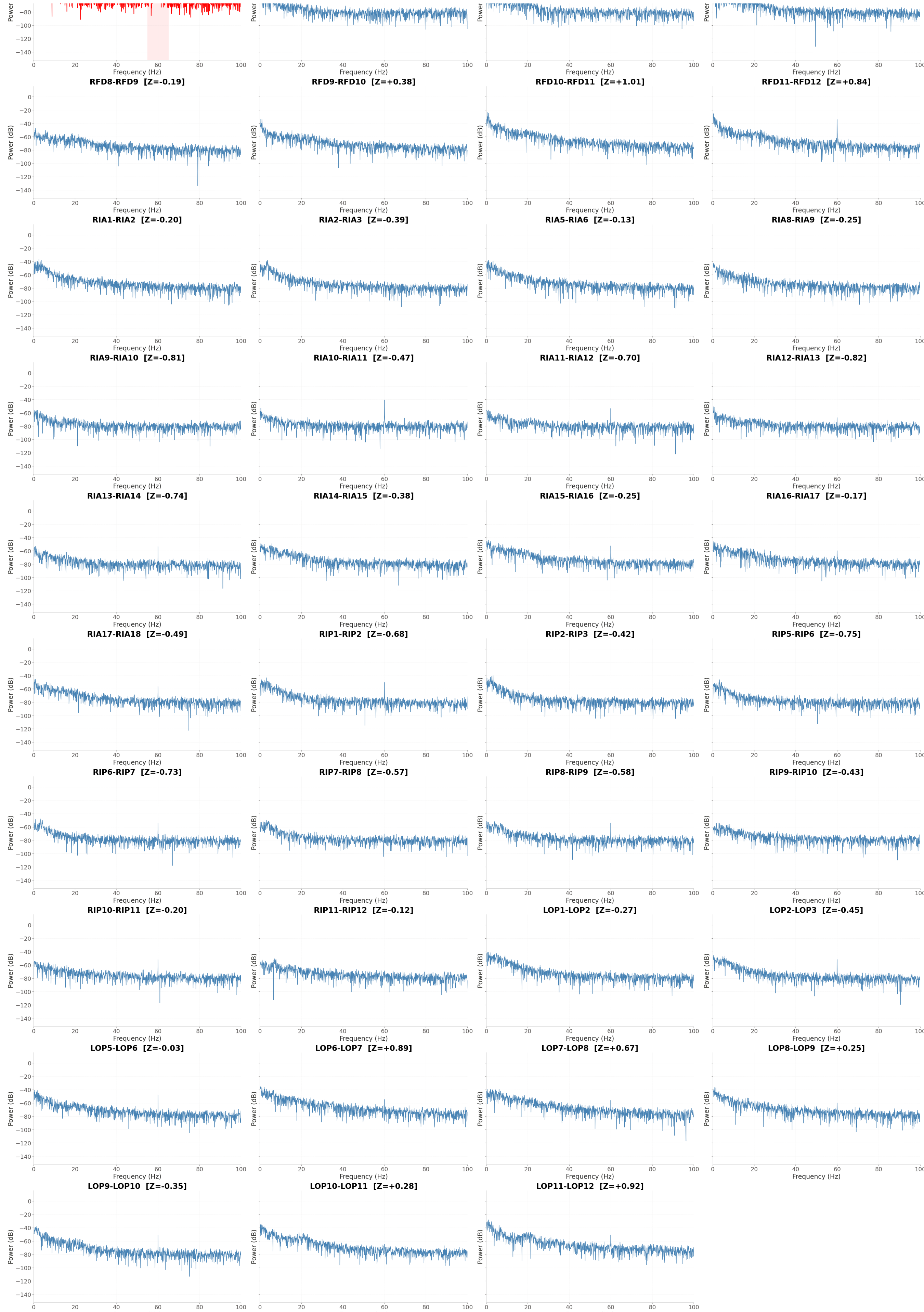


Frequency (Hz)
Power (dB)
RA4-RA5 [Z=-0.35]
RA5-RA6 [Z=-0.55]
RA6-RA7 [Z=-0.69]
RA7-RA8 [Z=-0.84]
RA8-RA9 [Z=-0.68]
RA9-RA10 [Z=-0.83]
RA10-RA11 [Z=+1.05]
RA11-RA12 [Z=+1.79]
RB1-RB2 [Z=+1.99]
RB2-RB3 [Z=+2.19]
RB3-RB4 [Z=+2.30]
RB4-RB5 [Z=+2.36]
RB5-RB6 [Z=-0.55]
RB6-RB7 [Z=-0.65]
RB7-RB8 [Z=-0.52]
RB8-RB9 [Z=+5.29]
RB9-RB10 [Z=+5.28]
RB10-RB11 [Z=-0.70]
RB11-RB12 [Z=-0.51]
LIA1-LIA2 [Z=-0.62]
LIA2-LIA3 [Z=-0.67]
LIA5-LIA6 [Z=+0.01]
LIA8-LIA9 [Z=+0.29]
LIA9-LIA10 [Z=-0.57]
LIA10-LIA11 [Z=-0.85]
LIA11-LIA12 [Z=-0.66]
LIA12-LIA13 [Z=-0.84]
LIA13-LIA14 [Z=-0.91]
LIA14-LIA15 [Z=+0.06]
LIA15-LIA16 [Z=+0.44]
LIA16-LIA17 [Z=+0.53]
LIA17-LIA18 [Z=+0.40]
LR1-LR2 [Z=-0.65]
LR2-LR3 [Z=-0.73]
LR3-LR4 [Z=-0.53]
LR4-LR5 [Z=-0.81]
LR5-LR6 [Z=-0.72]
LR6-LR7 [Z=-0.74]
LR7-LR8 [Z=-0.26]
LR8-LR9 [Z=+0.19]



LR9-LR10 [Z=+0.37]
LR10-LR11 [Z=-0.14]
LR11-LR12 [Z=+0.11]
ROA1-ROA2 [Z=-0.08]
ROA2-ROA3 [Z=-0.58]
ROA3-ROA4 [Z=-0.62]
ROA4-ROA5 [Z=-0.31]
ROA5-ROA6 [Z=-0.30]
ROA6-ROA7 [Z=+0.66]
ROA7-ROA8 [Z=+0.51]
ROA8-ROA9 [Z=+0.19]
ROA9-ROA10 [Z=-0.58]
ROA10-ROA11 [Z=-0.18]
ROA11-ROA12 [Z=+0.03]
ROP1-ROP2 [Z=-0.27]
ROP2-ROP3 [Z=-0.02]
ROP3-ROP4 [Z=-0.26]
ROP4-ROP5 [Z=-0.20]
ROP5-ROP6 [Z=-0.10]
ROP6-ROP7 [Z=-0.43]
ROP7-ROP8 [Z=-0.59]
ROP8-ROP9 [Z=-0.40]
ROP9-ROP10 [Z=-0.24]
ROP10-ROP11 [Z=-0.06]
ROP11-ROP12 [Z=+0.16]
RP1-RP2 [Z=-0.08]
RP2-RP3 [Z=+0.29]
RP3-RP4 [Z=+0.99]
RP4-RP5 [Z=+0.91]
RP5-RP6 [Z=-0.02]
RP6-RP7 [Z=-0.06]
RP7-RP8 [Z=-0.66]
RC1-RC2 [Z=+3.07]
RC2-RC3 [Z=+2.64]
RC3-RC4 [Z=+3.38]
RC4-RC5 [Z=+0.21]
Power (dB)
Frequency (Hz)


RC5-RC6 [Z=-0.16]
RC6-RC7 [Z=-0.56]
RC7-RC8 [Z=-0.64]
RC8-RC9 [Z=-0.73]
RC9-RC10 [Z=-1.01]
RC10-RC11 [Z=-0.73]
RC11-RC12 [Z=-0.22]
LFD1-LFD2 [Z=+0.44]
LFD2-LFD3 [Z=+0.22]
LFD3-LFD4 [Z=-0.87]
LFD4-LFD5 [Z=-0.76]
LFD5-LFD6 [Z=+0.23]
LFD6-LFD7 [Z=+0.00]
LFD7-LFD8 [Z=-0.10]
LFD8-LFD9 [Z=+0.34]
LFD9-LFD10 [Z=+0.09]
LIP1-LIP2 [Z=-0.17]
LIP2-LIP3 [Z=-0.33]
LIP3-LIP4 [Z=-0.28]
LIP4-LIP5 [Z=-0.21]
LIP5-LIP6 [Z=+0.52]
LIP6-LIP7 [Z=+0.54]
LIP7-LIP8 [Z=+0.41]
LIP8-LIP9 [Z=+0.43]
LIP9-LIP10 [Z=+0.60]
RD1-RD2 [Z=+3.16]
RD2-RD3 [Z=+2.67]
RD5-RD6 [Z=+0.44]
RD8-RD9 [Z=-0.59]
RD9-RD10 [Z=-0.70]
RD10-RD11 [Z=-0.68]
RD11-RD12 [Z=-1.03]
RD12-RD13 [Z=-0.83]
RD13-RD14 [Z=-0.69]
RD14-RD15 [Z=-0.61]
RFD1-RFD2 [Z=+3.48]
RFD2-RFD3 [Z=+3.43]
RFD5-RFD6 [Z=-1.03]
RFD6-RFD7 [Z=-0.83]
RFD7-RFD8 [Z=-0.79]
Frequency (Hz)
Power (dB)


RFD8-RFD9 [Z=-0.19]
RFD9-RFD10 [Z=+0.38]
RFD10-RFD11 [Z=+1.01]
RFD11-RFD12 [Z=+0.84]
RIA1-RIA2 [Z=-0.20]
RIA2-RIA3 [Z=-0.39]
RIA5-RIA6 [Z=-0.13]
RIA8-RIA9 [Z=-0.25]
RIA9-RIA10 [Z=-0.81]
RIA10-RIA11 [Z=-0.47]
RIA11-RIA12 [Z=-0.70]
RIA12-RIA13 [Z=-0.82]
RIA13-RIA14 [Z=-0.74]
RIA14-RIA15 [Z=-0.38]
RIA15-RIA16 [Z=-0.25]
RIA16-RIA17 [Z=-0.17]
RIA17-RIA18 [Z=-0.49]
RIP1-RIP2 [Z=-0.68]
RIP2-RIP3 [Z=-0.42]
RIP5-RIP6 [Z=-0.75]
RIP6-RIP7 [Z=-0.73]
RIP7-RIP8 [Z=-0.57]
RIP8-RIP9 [Z=-0.58]
RIP9-RIP10 [Z=-0.43]
RIP10-RIP11 [Z=-0.20]
RIP11-RIP12 [Z=-0.12]
LOP1-LOP2 [Z=-0.27]
LOP2-LOP3 [Z=-0.45]
LOP5-LOP6 [Z=-0.03]
LOP6-LOP7 [Z=+0.89]
LOP7-LOP8 [Z=+0.67]
LOP8-LOP9 [Z=+0.25]
LOP9-LOP10 [Z=-0.35]
LOP10-LOP11 [Z=+0.28]
LOP11-LOP12 [Z=+0.92]
Power (dB)
Frequency (Hz)

[]
Extremely low-power channel may be dead, disconnected, saturated, or otherwise not ca
rrying normal signal.